

# Cosmology HHT-BAO Locking Report

(Revision 1V9)

## Executive Summary

Application of Hilbert-Huang Transform (HHT) to SN residuals and subsequently test 'locking' to BAO effective scales.

New analyses (energy sweep, locking heatmaps, sliding- $\tau$  windows, and frequency-gated tests) consolidate that IMF2 carries a robust, statistically significant BAO-linked signal, while IMF1/IMF3 exhibit high but broadly shared co-variation that does not alter the IMF2 conclusion.

## Data & Method (brief)

**Signals:** SN residuals vs  $\log(1+z)$ ; BAO observables  $DM/r_d$  and  $DH/r_d$ .

**Decomposition:** EMD/EEMD/CEEMDAN; instantaneous frequency via Hilbert phase; uniform  $\tau$ -grid with cubic spline.

**Statistics:** Pearson  $r$  for locking, IMF2 energy ratio, phase-randomized surrogates ( $n \approx 1000$  per job), BH-FDR across cells/windows.

**Grids:** pads  $\in \{0.08, 0.10, 0.12, 0.15\}$ , ngrid  $\in \{192, 256, 320, 384\}$ ; seeds  $\in \{3, 7, 11, 23\}$ .

## Key Findings (1V9)

- IMF2 is consistently strongest: median  $r \approx 0.973$  at (pad=0.15, ngrid=384) with  $q \leq 0.0015$  across methods; IMF2 energy  $p \approx 0.0015$  (median across seeds).
- Sliding- $\tau$  windows (IMF2, CEEMDAN): 17/18 windows significant at  $q \leq 0.10$ ; peak  $\tau$ -centers  $\approx 0.615\text{--}0.815$  ( $z \approx 0.85\text{--}1.26$ ).
- Method-agnostic: EMD/EEMD/CEEMDAN agree on top cells; 'strong' ensembles (more trials/noise) are numerically identical to baseline.
- Adjacent modes: IMF1 (18/18) and IMF3 (16/18) windows significant, but frequency-gated tests (15–85% and 45–55% bands) change  $r$  by  $\leq 0.006$ , indicating multi-scale covariance rather than IMF2 leakage.

## Recommended Defaults

**Method:** CEEMDAN (EEMD/EMD equivalent here)

**Grid:** pad=0.15, ngrid=384

**IMF:** IMF2 for primary metrics (locking r, energy ratio)

**Surrogates:** ns=1000; BH-FDR across grid/windows

**Progress:** --progress auto (rich/tqdm)

### Consensus best cells (energy + locking):

| Method  | pad  | ngrid | IMF2 energy p<br>(med) | Locking r (med) |
|---------|------|-------|------------------------|-----------------|
| EMD     | 0.15 | 384   | 0.0015                 | 0.973           |
| EEMD    | 0.15 | 384   | 0.0015                 | 0.973           |
| CEEMDAN | 0.15 | 384   | 0.0015                 | 0.973           |

## Robustness

- Seed stability: near-zero IQR at top cells; results invariant across seeds (3,7, 11,23).
- Ensemble strength: ‘strong’ EEMD/CEEMDAN equal to baseline ( $\Delta r=0$ ,  $\Delta \text{IQR}=0$  across grid).
- Frequency-gating: IMF1/IMF3  $\Delta r \approx 0$  to  $-0.006$  as gate widens; IMF2 unchanged.

## Next Steps

- Export paper figures: energy heatmap, locking heatmap, sliding- $\tau$  curve, and specificity panel.
- Redshift-sliced scans to localize contributions and rule out edge artifacts.
- Optional: distance-based gating ( $|k_{\text{imf}} - k_2|/|k_2| < \delta$ ) for an even stricter specificity test.

*Baseline context: Cosmology HHT-BAO Locking Report (Revision 1V8).*